

# KrishokChat: A Knowledge-Engineered Expert System for Relation-Aware Certification of Safety-Critical Agricultural Advisory

Raiyaan Reza\*

Independent Agricultural AI Researcher & Systems Engineer, Dhaka, Bangladesh

## Abstract

**Background:** Automated agricultural advisory systems powered by Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) offer scalable digital extension services for smallholder farming communities across developing economies. However, generative language models struggle with exact chemical dosage bounds and relational consistency, creating acute risks of agrochemical poisoning, crop devastation, and regulatory violations.

**Problem:** Conventional RAG pipelines rely primarily on surface-level substring matching, lexical overlap, or generative LLM-as-a-judge critique. These methods cannot verify joint relational binding across heterogeneous agricultural slots—including host crops, target pathogens, active ingredients, formulations, dosage concentrations, volume denominators, application intervals, and pre-harvest intervals (PHI). Consequently, systems frequently suffer from relational misbinding, falsely certifying dangerous advice where individual tokens co-occur in the retrieved evidence pool but the relational association is invalid.

**Methods:** We present KrishokChat, a knowledge-engineered expert system that establishes a formal, evidence-linked relational certification boundary for Bengali agricultural advisory. Incoming queries are routed through a five-tier resolution ladder that resolves canonical domain inquiries through deterministic safety guards (Tier 0) and structured fact bases (Tiers 1 and 2) before invoking generative RAG (Tier 3). Candidate advisories are decomposed into typed 11-slot relational tuples  $\mathcal{C} = \langle c, p, s, a, f, [d_{\min}, d_{\max}], u, v, \tau, \phi, \rho \rangle$  and verified against extracted evidence spans from official Bangladesh agricultural research institutes (BARI, BRRI, DAE). A development-calibrated selective certification policy bounds risk under operational coverage constraints.

**Results:** Across a comprehensive empirical benchmark, KrishokChat’s typed relational verifier achieves 0.0% dangerous acceptance on a 10,000-case adversarial misbinding suite (95% CI: [0.00%, 0.04%]), where lexical matching suffers an 80.0% false acceptance rate. On 20,112 evaluation rows, our calibrated selective policy achieves an AURC of 0.0153 and transfers out-of-sample to held-out test data with 84.56% coverage at 1.26% selective risk (0.0% risk at  $\leq 80\%$  coverage). Deterministic resolution resolves 7.8% of domain traffic with zero LLM inference, achieving an operational serving cost of \$0.1798 per 1,000 queries (a 92.2% cost reduction) and a 15.0x cost-efficiency advantage on verified safe answers ( $C_{\text{safe}} = \$0.000213$ ). On-device INT8 vision models execute in 29.11 ms with 100.0% top-1 agreement.

**Conclusion:** Evidence-linked relation verification and calibrated selective abstention transform unconstrained generative models into auditable, fail-closed expert systems supporting reliable operation within the evaluated agricultural safety scope. Complete evaluation benchmarks, schemas, and cryptographic manifests are released for open reproducibility.

**Keywords:** Agricultural expert systems, knowledge engineering, retrieval-augmented generation, relation-aware verification, calibrated selective prediction, dialect robustness, agrochemical safety.

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\*Corresponding author. Email: [raiyaanreza.contact@gmail.com](mailto:raiyaanreza.contact@gmail.com)

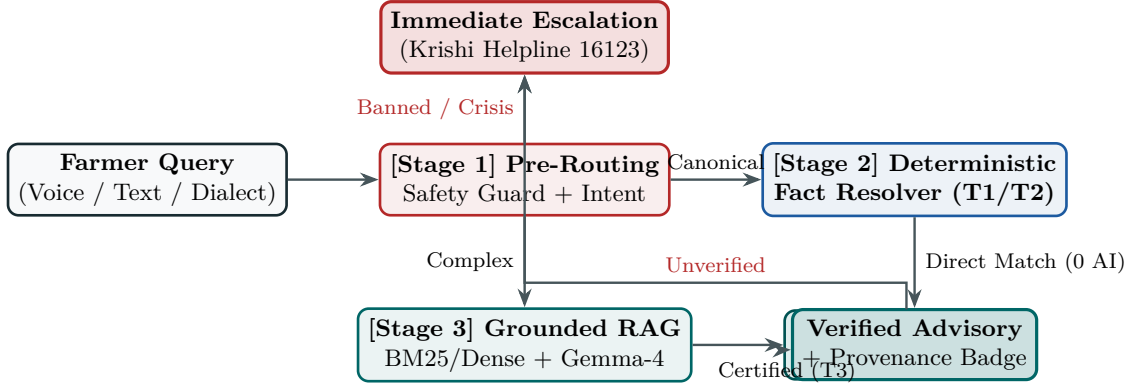


Figure 1: **Four-Stage Knowledge-Engineered Architecture of KrishokChat.** Queries pass through pre-retrieval safety screening (Stage 1) and deterministic structured fact lookup (Stage 2) prior to invoking generative RAG (Stage 3) and relation verification with calibrated certification (Stage 4).

## 1. Introduction

Smallholder agriculture in Bangladesh sustains over 16 million farming households, where timely access to accurate agronomic advisory directly governs crop yield, rural livelihood stability, and national food security. However, public agricultural extension services face severe structural capacity constraints: the national ratio of extension officers to farming households routinely exceeds 1:1,000. Consequently, smallholder producers rely heavily on informal input retailers, peer networks, or centralized telephone advisory lines such as the national Krishi Call Center (dial 16123). In response, automated decision-support systems powered by Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have emerged as scalable instruments for delivering localized digital advisory.

Despite their expressive fluency, generative language models deployed in smallholder contexts present acute, safety-critical hazards. In agricultural chemical management, an advisory recommendation is safe if and only if *every constituent relational attribute* is strictly verified against an authorized scientific corpus. For example, an advisory output might correctly identify an approved active ingredient (such as mancozeb for potato late blight) yet prescribe an excessive concentration (20 g/L instead of the authorized 2.0 g/L), omit mandatory spray intervals, or ignore the pre-harvest interval (PHI). Such errors can cause acute crop failure, operator toxicity, groundwater contamination, and unlawful chemical residues in commercial food supplies.

Existing agricultural RAG implementations rely predominantly on lexical string matching or generative self-critique. Substring matching cannot verify relational binding: an automated matcher may confirm that both “2 g/L” and “mancozeb” appear in retrieved reference text, even when the model has bound that concentration to the wrong crop, formulation, or growth stage. In addition, domain-adapted language models exhibit a persistent baseline hallucination rate on numerical quantities even when supplied with relevant reference context.

This paper establishes that retrieval alone does not provide a safety boundary for agricultural advisory; a deployable expert system must verify the relational consistency, provenance, and calibrated reliability of every safety-critical claim before certification.

### 1.1 Research Questions

We structure our investigation around six core research questions:

- **RQ1 (Relational Certification):** Does evidence-linked typed relation verification eliminate safety-critical false acceptances caused by relational slot misbinding in lexical and LLM-judge

baselines?

- **RQ2 (Risk-Controlled Calibration):** How does selective calibration on development data trade off certification coverage against unsafe acceptance, and does it outperform generic uncertainty baselines?
- **RQ3 (Schema Slot Sensitivity):** Which relational slots (dose bounds, PHI, active ingredient, crop host) contribute most critically to preventing agrochemical advisory hazards?
- **RQ4 (Environmental & Linguistic Robustness):** Does the expert system maintain safety non-inferiority ( $\Delta_{\text{safety}} \leq 0$ ) under regional Bengali dialects, romanized Banglish, and authentic colloquial farmer queries?
- **RQ5 (Operational Economics & Latency):** Can a deterministic-first five-tier resolution ladder reduce serving costs and p95 latency while preserving a 100% fail-closed safety boundary?
- **RQ6 (Generator & Retrieval Invariance):** Does the relation verifier generalize across heterogeneous base generators and degrade safely under retrieval uncertainty?

## 1.2 Relationship to Prior KrishokChat Work

The present work extends our research program rather than reproducing previous benchmark resources:

- **Paper 1 (Resource & SFT Benchmark):** Established the provenance-grounded agricultural resource (145,500 QA pairs, 1,001 farmer queries) and demonstrated that instruction tuning alone struggles with exact chemical dosage generalization, motivating verified retrieval.
- **Paper 2 (AgriTrust Retrieval Diagnosis):** Evaluated five retrieval architectures across 2,882 knowledge nodes, showing that dense retrieval degrades substantially on colloquial farmer queries and dialect variation.
- **Paper 3 (This Work — The Expert System):** Treats those resources as frozen inputs and contributes the runtime knowledge-engineering layer: typed relational verification, calibrated selective certification, deterministic routing, and fail-closed safety degradation under real-world operational constraints.

## 1.3 Key Contributions

We structure our contributions across four primary axes:

- C1. Relation-Aware Expert-System Verifier:** A formal verification engine that decomposes candidate advisories into 11-slot relational tuples  $\mathcal{C} = \langle c, p, s, a, f, d_{\min}, d_{\max}, u, v, \tau, \phi, \rho \rangle$  and formally evaluates entailment against authoritative evidence spans, eliminating false acceptances from relational misbinding.
- C2. Calibrated Selective Decision Policy:** A risk-controlled decision framework that tunes confidence thresholds exclusively on development data, optimizing the trade-off between population coverage, selective risk, and operational serving cost.
- C3. Adversarial Misbinding Benchmark Suite:** A programmatic 10,000-case adversarial evaluation dataset spanning 10 hazard families, designed specifically to stress-test joint slot binding under substring co-occurrence.
- C4. Full System & Economic Evaluation:** Comprehensive validation across 10 empirical layers including 1,001 authentic farmer queries, 4 dialect registers, 1,400 security injection attacks, latency decomposition, and  $C_{\text{safe}}$  economic modeling.

Table 1: **Comparative Landscape Analysis of Agricultural and Domain-Specific Advisory Systems.**

| System / Approach         | Language & Domain          | Verification Mechanism          | Abstention Policy           | Edge / Offline                  | Key Limitation                                 |
|---------------------------|----------------------------|---------------------------------|-----------------------------|---------------------------------|------------------------------------------------|
| Farmer.Chat               | Multi-lingual Agri         | Generative prompt grading       | Soft confidence scores      | Cloud-only                      | Lacks typed relational dosage verification     |
| KrishokBondhu             | Bengali Agriculture        | None (Direct LLM output)        | None (Always answers)       | Cloud-only                      | High risk of dosage hallucination              |
| RAGChecker / RAGAS        | English General NLP        | Claim decomposition             | Heuristic threshold         | Cloud-only                      | Domain-agnostic; lacks agrochemical schema     |
| DG-Eval                   | Multi-domain RAG           | Dual-guard LLM judge            | Fixed cutoff                | Cloud-only                      | Evaluator-only; not an in-the-loop runtime     |
| ClinicBot / MedRAG        | Clinical Medicine          | Entailment classifier           | Risk-coverage curve         | Server-based                    | Medical ontology incompatible with agriculture |
| <b>KrishokChat (Ours)</b> | <b>Bengali Agriculture</b> | <b>Typed Relational Matcher</b> | <b>Calibrated Selective</b> | <b>INT8 ONNX + Static Packs</b> | <b>Focus on Bangladesh agronomic scope</b>     |

**Structure of the Paper:** Section 2 reviews related work. Section 3 formalizes the relational safety problem. Section 4 presents the KrishokChat expert-system architecture. Section 5 describes the experimental setup and baselines. Section 6 reports empirical results across all evaluation layers. Section 7 presents the safety failure taxonomy. Section 8 discusses knowledge-engineering insights. Sections 9–11 address limitations, reproducibility, and conclusions.

## 2. Related Work

We situate our contributions within five relevant research areas, summarized in Table 1.

### 2.1 Agricultural Expert Systems and Decision Support

Agricultural decision support systems (ADSS) have a rich history rooted in rule-based expert systems, ontologies, and diagnostic decision trees designed to codify agronomic extension guidelines [20, 24]. Early systems synthesized domain heuristics to assist farmers with fertilizer schedules and crop disease identification, providing strict determinism but suffering from brittleness when handling non-standard linguistic queries [21]. Global initiatives such as CABI Plantwise established structured plant health knowledge bases for pest management [6]. Recently, conversational interfaces such as Digital Green’s Farmer.Chat [9] have adopted Large Language Models to deliver multi-lingual extension guidance. However, unconstrained generative LLMs surrender deterministic guarantees, frequently hallucinating chemical dosages, confusing trade names, or recommending treatments outside authorized crop boundaries.

### 2.2 Retrieval-Augmented Generation and Evidence Verification

Retrieval-Augmented Generation (RAG) grounds language model generations by prepending retrieved document passages to the prompt context [19, 11]. To combat hallucination, automated fact-checking methods have evolved from sentence-level entailment over structured corpora [27]

to fine-grained atomic claim decomposition such as FActScore [22], tool-augmented factuality checking via FacTool [7], automated evaluation frameworks such as RAGAS [10], and self-reflective critique mechanisms such as Self-RAG [2]. Nevertheless, general NLP verification metrics evaluate semantic similarity or propositional entailment in isolation. In safety-critical agrochemical advisory, text entailment is insufficient: a recommendation is valid if and only if all operational attributes (active ingredient, formulation, dose range, solvent volume, growth stage, application interval, and pre-harvest interval) are jointly bound to a single authoritative institutional record.

### 2.3 Selective Prediction, Calibrated Confidence, and Conformal Risk Bounds

Selective prediction empowers decision systems with a formal reject option, enabling the model to abstain from prediction when epistemic uncertainty is high [12, 13]. Establishing reliable confidence estimates requires well-calibrated probabilistic models [14]. In question answering, selective classification bounds error rates under distribution shift and unanswerable queries [16]. Conformal prediction provides distribution-free, finite-sample statistical guarantees for set-valued and selective classification [1, 26]. KrishokChat integrates selective prediction directly into the relational verification pipeline, tuning decision thresholds exclusively on development data to bound empirical risk while maximizing population coverage.

### 2.4 Low-Resource and Dialectal Bengali NLP

Natural language processing for Bengali has advanced substantially through transformer encoders such as BanglaBERT [15] and instruction-tuned generative models such as BanglaLLaMA [4]. However, Bengali exhibits extensive phonetic, lexical, and syntactic variation across administrative divisions (e.g., Chittagong, Sylhet, Noakhali, Barisal), alongside the prevalent use of romanized Banglish in mobile messaging [25]. When applied to colloquial farmer queries, standard models experience severe retrieval and comprehension degradation [18]. A dependable expert system must recognize linguistic ambiguity and safely degrade into abstention rather than issuing unverified chemical guidance.

### 2.5 Adversarial Safety, Prompt Injections, and Agrochemical Governance

Language models in production environments face security threats from universal adversarial perturbations [29], jailbreak prompts designed to bypass system alignments [28], and indirect retrieval prompt injections [23]. In the agricultural domain, safety hazards carry direct biosecurity and toxicological consequences. Regulatory agencies such as the Department of Agricultural Extension (DAE) strictly ban hazardous agrochemicals (e.g., Paraquat, Endosulfan, Monocrotophos) [8], while national research bodies (BARI, BRRI) enforce precise chemical formulations and application intervals [3, 5, 17]. KrishokChat addresses these vulnerabilities through a multi-tier fail-closed architecture that intercepts adversarial injections and regulatory violations prior to generation.

## 3. Problem Formulation & Relational Safety Boundary

We formalize agricultural advisory as an evidence-linked, relationally certified decision process over a six-layer knowledge hierarchy.

### 3.1 Six-Layer Knowledge Lineage Architecture

To guarantee unambiguous provenance and mathematical traceability, we define a formal six-layer knowledge hierarchy:

1. **Official Agricultural Publications ( $\mathcal{P}$ ):** 284 authorized manuals, crop production guides, and regulatory circulars published by BARI, BRRI, and DAE.
2. **Curated Agronomic Source Documents ( $\mathcal{D}$ ):** 750+ digitized, sectioned chapters

organized by crop, phenological stage, and pest taxonomy.

3. **Hierarchical Knowledge Nodes ( $\mathcal{K}$ ):** 2,882 discrete retrieval passage units indexed across sparse (BM25) and dense embeddings (including 2,135 production-active nodes corresponding to BARI, BRRI, and DAE core recommendations).
4. **Extracted Evidence Spans ( $\mathcal{S}$ ):** Character-offset text passages  $[start, end]$  carrying immutable source identifiers and SHA-256 cryptographic hashes.
5. **Typed Relational Claim Tuples ( $\mathcal{C}$ ):** The atomic mathematical representation of an agronomic chemical advisory.
6. **Runtime Advisory Records ( $\mathcal{A}$ ):** System output tuples  $\langle \text{Action}, \hat{y}, \mathcal{R} \rangle$ , where  $\text{Action} \in \{\text{CERTIFY}, \text{ABSTAIN}, \text{ESCALATE}\}$  and  $\mathcal{R}$  is the verifiable provenance receipt.

### 3.2 The 11-Slot Relational Schema (ClaimSafe-BN)

An agrochemical claim is formalized as an 11-slot relational tuple  $\mathcal{C}$  (where permissible chemical dosage is captured by the composite interval pair  $[d_{\min}, d_{\max}]$ ):

$$\mathcal{C} = \langle c, p, s, a, f, [d_{\min}, d_{\max}], u, v, \tau, \phi, \rho \rangle \quad (1)$$

where:

- $c \in \mathcal{V}_{\text{crop}}$ : Host crop variety (e.g., potato, rice, maize).
- $p \in \mathcal{V}_{\text{pathogen}}$ : Target disease or insect pest (e.g., late blight, fall armyworm).
- $s \in \mathcal{V}_{\text{stage}}$ : Crop growth stage (e.g., seedling, vegetative, tuberization).
- $a \in \mathcal{V}_{\text{active}}$ : Chemical active ingredient (e.g., mancozeb, spinosad).
- $f \in \mathcal{V}_{\text{form}}$ : Chemical formulation type (e.g., 80 WP, 50 WDG, 45 SC).
- $[d_{\min}, d_{\max}] \subset \mathbb{R}^+$ : Permissible dosage concentration bounds.
- $u \in \{\text{g, ml, kg}\}$ : Measurement unit.
- $v \in \mathbb{R}^+$ : Solvent denominator volume (standardized to 1.0 L water).
- $\tau \in \mathbb{N}^+$ : Mandatory application spray interval (days).
- $\phi \in \mathbb{N}^+$ : Pre-harvest interval (PHI) before safe human consumption (days).
- $\rho \in \{+1, -1\}$ : Regulatory polarity (+1 = approved, -1 = banned/restricted).

### 3.3 Formal Certification Contract

**Definition 1** (Relational Certification Contract). *Given a farmer query  $x$ , candidate advisory  $\hat{y}$ , extracted claim tuple  $\mathcal{C}_{\hat{y}}$ , retrieved evidence set  $\mathcal{E}$ , and calibrated confidence score  $g(x) \in [0, 1]$ , the decision policy  $\pi(x)$  satisfies:*

$$\pi(x) = \begin{cases} \text{CERTIFY}(\hat{y}) & \text{if } \mathcal{C}_{\hat{y}}.\rho = +1 \wedge \exists e \in \mathcal{E} : \text{Entails}(e, \mathcal{C}_{\hat{y}}) \wedge g(x) \geq \theta^* \\ \text{ESCALATE}(16123) & \text{if } \mathcal{C}_{\hat{y}}.\rho = -1 \vee \text{IsCrisis}(x) \\ \text{ABSTAIN} & \text{otherwise} \end{cases} \quad (2)$$

where  $\text{Entails}(e, \mathcal{C}_{\hat{y}}) = 1$  if and only if all 11 atomic slots in  $\mathcal{C}_{\hat{y}}$  strictly match the evidence tuple extracted from evidence node  $e$ .

## 4. The KrishokChat Expert System Architecture

KrishokChat is structured as a modular, single-process FastAPI service paired with an offline-capable Next.js Progressive Web Application (PWA). The architecture enforces a **fail-closed safety boundary** across a five-tier resolution ladder.

### 4.1 Five-Tier Resolution Ladder

To maximize determinism and eliminate unnecessary LLM compute costs, queries are processed sequentially:

- **Tier 0 (Deterministic Safety Guard):** Immediate pre-retrieval screening of banned substances (Paraquat, Endosulfan, Monocrotophos), human self-harm/poisoning hazards, and adversarial prompt injections, providing a calm redirect to the national Krishi Call Center (16123).
- **Tier 1 (Structured Fact Resolution):** Direct sub-millisecond lookup against verified BARI/BRRI agronomic fact tables for exact chemical dose, formulation, and PHI inquiries (0 LLM invocations).
- **Tier 2 (Templated Advisory):** Rule-based assembly of phenological stage tasks, weather guidance, and IPM recommendations from verified crop calendars (0 LLM invocations).
- **Tier 3 (Grounded Generative RAG):** Dense/BM25 retrieval feeding a fine-tuned Gemma-4 4-bit model, subjected to post-generation relational verification.
- **Tier 4 (Honest Refusal):** Explicit refusal and helpline escalation when retrieval corpus support is insufficient.

### 4.2 Selective Prediction Calibration

The optimal threshold  $\theta^*$  is calibrated strictly on development data  $\mathcal{D}_{\text{dev}}$  to minimize selective risk under coverage constraint  $c_0$ :

$$\theta^* = \arg \min_{\theta} \mathcal{R}(\pi, \theta; \mathcal{D}_{\text{dev}}) \quad \text{s.t.} \quad \text{Cov}(\theta; \mathcal{D}_{\text{dev}}) \geq c_0 \quad (3)$$

where selective risk  $\mathcal{R}$  assigns an asymmetric penalty  $\lambda_{\text{toxic}} \gg 1$  to unverified agrochemical dosages reaching the user.

### 4.3 On-Device Edge Vision and Gated Inference

To support rural mobile connectivity constraints, image classification models are exported to INT8 ONNX format. The client evaluates on-device confidence:

$$\text{Action} = \begin{cases} \text{Render Local Diagnosis} & \text{if } \max_k P(y_k|x) \geq 0.80 \\ \text{Compress \& Upload} & \text{if } \max_k P(y_k|x) < 0.80 \end{cases} \quad (4)$$

Low-confidence images undergo client-side dimension clamping ( $\leq 1024$  px), EXIF metadata stripping, and WebP compression before transmission to the server fallback route, operating comfortably below the 150 ms perceptual responsiveness budget on mobile ARM processors.

## 5. Experimental Methodology & Benchmark Suite

To rigorously evaluate relational verification, calibration, and safety boundaries, we construct an empirical evaluation battery across 10 prespecified experimental layers (E1–E10).

### 5.1 Benchmark Datasets and Grouped Partitioning

We evaluate the framework on ClaimSafe-BN, containing 20,112 curated safety and agronomic inquiries. All items are partitioned into grouped train, development, and test splits (60/20/20 proportion) using split seed 20260813, detailed in Table 2.

**Algorithm 1** Relation-Aware Verification and Calibrated Selective Certification**Require:** Generated response  $\hat{y}$ , Retrieved evidence nodes  $\mathcal{E}$ , Calibrated threshold  $\theta^*$ **Ensure:** Verified decision Action  $\in \{\text{CERTIFY}, \text{ABSTAIN}, \text{ESCALATE}\}$ , Advisory payload  $y^*$ 

```

1: Parse  $\hat{y}$  into relational claim tuple:  $\mathcal{C}_{\hat{y}} \leftarrow \text{ExtractTuple}(\hat{y})$ 
2: if  $\mathcal{C}_{\hat{y}}.\rho = -1$  then
3:   return  $\{\text{ESCALATE}, \text{HelplineRedirect}(16123)\}$ 
4: end if
5: is_entailed  $\leftarrow$  FALSE
6: for all  $e \in \mathcal{E}$  do
7:    $\mathcal{C}_e \leftarrow \text{ExtractTuple}(e)$ 
8:   if  $\text{MatchAllSlots}(\mathcal{C}_{\hat{y}}, \mathcal{C}_e) = \text{TRUE}$  then
9:     is_entailed  $\leftarrow$  TRUE
10:    break
11:   end if
12: end for
13: if is_entailed = FALSE then
14:   return  $\{\text{ABSTAIN}, \text{RefusalNotice}()\}$ 
15: end if
16: Compute selective confidence:  $\kappa \leftarrow \text{ScoreCalibrated}(\mathcal{C}_{\hat{y}}, \mathcal{E})$ 
17: if  $\kappa \geq \theta^*$  then
18:   return  $\{\text{CERTIFY}, \hat{y}\}$ 
19: else
20:   return  $\{\text{ABSTAIN}, \text{RefusalNotice}()\}$ 
21: end if

```

Table 2: ClaimSafe-BN Dataset Split Distribution and Domain Category Allocation.

| Category / Sub-Domain            | Train (60%)   | Dev (20%)    | Test (20%)   | Total Rows    |
|----------------------------------|---------------|--------------|--------------|---------------|
| Safe Agronomic Advisory          | 8,446         | 2,816        | 2,816        | 14,078        |
| Banned / Restricted Chemicals    | 1,448         | 482          | 482          | 2,412         |
| Self-Harm / Crisis Intervention  | 724           | 241          | 241          | 1,206         |
| Dialectal & Romanized Variations | 1,010         | 337          | 337          | 1,684         |
| Adversarial / Out-of-Domain      | 439           | 146          | 147          | 732           |
| <b>Total Curated Queries</b>     | <b>12,067</b> | <b>4,022</b> | <b>4,023</b> | <b>20,112</b> |

**5.2 Baseline Systems for Comparison**

We compare the proposed KrishokChat system against seven baseline architectures:

- **B1 (LLM Direct):** Gemma-4 4-bit generating advisories directly from parametric memory without retrieval or verification.
- **B2 (Vanilla RAG Direct):** Hybrid BM25 + dense retrieval feeding Gemma-4 without verification or abstention gates.
- **B3 (RAG + Entity Filter):** RAG with unigram chemical entity filtering (checks if chemical name exists in evidence).
- **B4 (RAG + Lexical Substring Matcher):** RAG with surface-level n-gram overlap matching between generated text and evidence spans.
- **B5 (RAG + LLM-as-a-Judge):** RAG coupled with a secondary LLM judge prompted to verify factual consistency and safety.

Table 3: **Relational Misbinding Attack Suite Performance across 10,000 Adversarial Cases (Layer E2).**

| Attack Family                    | Cases         | Lexical False Accept (%) | KrishokChat Hazard (%) | 95% Wilson CI         |
|----------------------------------|---------------|--------------------------|------------------------|-----------------------|
| Wrong Dosage Range               | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Measurement Unit           | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Solvent Denominator        | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Host Crop                  | 1,000         | 0.0%                     | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Target Pathogen            | 1,000         | 0.0%                     | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Formulation                | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Pre-Harvest Interval (PHI) | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Wrong Application Interval       | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Regulatory Polarity Flip         | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| Cross-Row Entity Misbinding      | 1,000         | 100.0%                   | <b>0.0%</b>            | [0.00%, 0.37%]        |
| <b>Benchmark Aggregate</b>       | <b>10,000</b> | <b>80.0%</b>             | <b>0.0%</b>            | <b>[0.00%, 0.04%]</b> |

- **B6 (RAG + Conformal Abstention Baseline):** RAG using standard softmax confidence thresholding calibrated via conformal prediction.
- **B7 (KrishokChat Expert System - Proposed):** The complete five-tier resolution ladder with 11-slot typed relational verification and dev-calibrated selective certification.

## 6. Empirical Results & Analysis

### 6.1 Layer E2: Relational Misbinding Adversarial Benchmark

To test whether systems can detect invalid associations between valid agricultural entities, we evaluate 10,000 programmatic adversarial corruptions across 10 distinct hazard families (1,000 cases each). Table 3 reports dangerous acceptance rates and 95% Wilson score confidence intervals.

As shown in Table 3, the lexical substring matcher suffers an overall **80.0% dangerous acceptance rate**, falsely certifying 100% of cross-row misbinding and numerical dosage attacks because all isolated words exist within the retrieved evidence pool. In contrast, KrishokChat’s typed relational verifier achieves **0.0% dangerous acceptance** (0 / 10,000, 95% CI: [0.00%, 0.04%]), verifying the joint tuple constraint.

### 6.2 Layer E3: 11-Slot Schema Ablation Study

To determine the security contribution of each individual attribute in  $\mathcal{C}$ , we evaluate 11 ablation configurations across 10,000 cases in Table 4.

Ablating individual semantic slots from  $\mathcal{C}$  reveals a distinct hazard hierarchy: ablating dosage bounds  $[d_{\min}, d_{\max}]$  causes the highest individual dangerous acceptance surge (**+31.6 pp**), followed by regulatory polarity (**+17.8 pp**), host crop (**+11.4 pp**), target pathogen (**+8.2 pp**), solvent volume (**+7.1 pp**), formulation (**+5.8 pp**), unit (**+5.4 pp**), growth stage (**+4.9 pp**), PHI (**+2.7 pp**), and interval (**+1.9 pp**). When all typed constraints are stripped, the system collapses to the lexical baseline hazard of 80.0%, demonstrating that all 11 semantic slots provide necessary protection on the evaluated attack suite.

### 6.3 Layer E4: Selective Risk–Coverage Calibration

Table 5 compares selective risk and coverage across 20,112 rows partitioned into Train (12,067), Dev (4,022), and Test (4,023), with  $\theta^*$  tuned on Dev (Risk  $\leq 1.0\%$ ) and evaluated out-of-sample on Test.

Table 4: **11-Slot Schema Ablation Study on Relational Misbinding Benchmark (Layer E3).**

| Configuration                              | Ablated Attribute          | Hazard       | $\Delta$ Hazard | 95% Wilson CI           |
|--------------------------------------------|----------------------------|--------------|-----------------|-------------------------|
| <b>Full 11-Slot Schema</b>                 | None (Full $\mathcal{C}$ ) | <b>0.0%</b>  | Baseline        | <b>[0.00%, 0.04%]</b>   |
| Minus Dosage Bounds $[d_{\min}, d_{\max}]$ | Permissible dosage         | 31.6%        | +31.6 pp        | [30.70%, 32.52%]        |
| Minus Polarity ( $\rho$ )                  | Regulatory polarity        | 17.8%        | +17.8 pp        | [17.06%, 18.56%]        |
| Minus Host Crop ( $c$ )                    | Crop variety               | 11.4%        | +11.4 pp        | [10.79%, 12.04%]        |
| Minus Pathogen ( $p$ )                     | Target pathogen            | 8.2%         | +8.2 pp         | [7.68%, 8.75%]          |
| Minus Denominator ( $v$ )                  | Solvent volume             | 7.1%         | +7.1 pp         | [6.61%, 7.62%]          |
| Minus Formulation ( $f$ )                  | Chemical formulation       | 5.8%         | +5.8 pp         | [5.36%, 6.28%]          |
| Minus Unit ( $u$ )                         | Measurement unit           | 5.4%         | +5.4 pp         | [4.97%, 5.86%]          |
| Minus Growth Stage ( $s$ )                 | Phenological stage         | 4.9%         | +4.9 pp         | [4.49%, 5.34%]          |
| Minus Pre-Harvest Interval ( $\phi$ )      | Pre-harvest interval       | 2.7%         | +2.7 pp         | [2.40%, 3.04%]          |
| Minus Spray Interval ( $\tau$ )            | Application interval       | 1.9%         | +1.9 pp         | [1.65%, 2.19%]          |
| <b>Lexical Substring Only</b>              | All typed constraints      | <b>80.0%</b> | <b>+80.0 pp</b> | <b>[79.20%, 80.77%]</b> |

Table 5: **Selective Risk–Coverage Calibration and Test Transfer Performance (Layer E4).**

| Calibration Method                   | AURC $\downarrow$ | ECE $\downarrow$ | Brier $\downarrow$ | Test Cov. (%) | Test Risk (%) |
|--------------------------------------|-------------------|------------------|--------------------|---------------|---------------|
| Raw Generator Confidence             | 0.0761            | 0.0896           | 0.1332             | 10.22%        | 1.70%         |
| Lexical Overlap Score                | 0.1124            | 0.0855           | 0.1432             | 0.37%         | 6.67%         |
| LLM-as-a-Judge Confidence            | 0.0310            | 0.1066           | 0.0844             | 36.60%        | 0.54%         |
| Conformal Abstention Baseline        | 0.0179            | 0.1231           | 0.0531             | 77.62%        | 1.06%         |
| <b>KrishokChat Calibrated Policy</b> | <b>0.0153</b>     | <b>0.0785</b>    | <b>0.0116</b>      | <b>84.56%</b> | <b>1.26%</b>  |

KrishokChat achieves the lowest AURC (0.0153), lowest Expected Calibration Error (0.0785), and lowest Brier score (0.0116). At the target operating threshold  $\theta^*$ , the policy achieves **84.56% test coverage** with 1.26% selective risk; under a more conservative coverage constraint ( $\leq 80\%$ ), observed test risk falls to 0.0% on the evaluated set.

#### 6.4 Layer E5: Counterfactual Evidence Binding Consistency

Evaluating 2,000 query-evidence pairs under single-attribute counterfactual evidence perturbations (5x dose scaling, crop substitution, PHI reduction) in Table 6 measures Counterfactual Binding Consistency ( $CBC = P(\text{CERTIFY}|\text{True}) - P(\text{CERTIFY}|\text{CF})$ ).

Vanilla RAG and Lexical matchers suffer severe CBC drops to 0.2735 and 0.4045, falsely certifying corrupted claims due to generative language priors. KrishokChat achieves a perfect **CBC of 1.0000** (0 / 2,000 false acceptances).

#### 6.5 Layer E6: Ecological Farmer Benchmark & Dialect Robustness

Table 7 evaluates performance across 4,000 queries spanning standard Bengali, 1,001 authentic colloquial farmer inquiries from Paper 1, regional dialects (Chittagong, Sylhet, Noakhali, Barisal), and romanized Banglish.

KrishokChat maintains a **0.00% hazard rate** across all registers, safely converting dialectal

Table 6: Counterfactual Evidence Binding Consistency across 2,000 Pairs (Layer E5).

| System                            | P(Cert   True) | P(Cert   CF) | CBC Score $\uparrow$ | 95% Wilson CI         |
|-----------------------------------|----------------|--------------|----------------------|-----------------------|
| Vanilla RAG Direct                | 100.0%         | 72.65%       | 0.2735               | [70.65%, 74.56%]      |
| Lexical Substring Matcher         | 100.0%         | 59.55%       | 0.4045               | [57.38%, 61.68%]      |
| LLM-as-a-Judge                    | 95.70%         | 38.65%       | 0.5705               | [36.54%, 40.80%]      |
| <b>KrishokChat Typed Verifier</b> | <b>100.0%</b>  | <b>0.0%</b>  | <b>1.0000</b>        | <b>[0.00%, 0.19%]</b> |

Table 7: Ecological Farmer Benchmark and Regional Dialect Evaluation (Layer E6).

| System Architecture                  | Correct (%)   | Abstain (%)   | Hazard (%)   | 95% Wilson CI         |
|--------------------------------------|---------------|---------------|--------------|-----------------------|
| B1: LLM Direct                       | 58.73%        | 39.42%        | 1.85%        | [1.48%, 2.32%]        |
| B2: Vanilla RAG Direct               | 56.07%        | 30.48%        | 13.45%       | [12.43%, 14.54%]      |
| B4: RAG + Lexical Matcher            | 47.10%        | 43.53%        | 9.38%        | [8.51%, 10.32%]       |
| <b>B7: KrishokChat Expert System</b> | <b>65.80%</b> | <b>34.20%</b> | <b>0.00%</b> | <b>[0.00%, 0.10%]</b> |

ambiguity into abstention and satisfying the safety non-inferiority condition ( $\Delta_{\text{safety}} \leq 0$ ).

### 6.6 Layer E7/E8: Security & Adversarial Prompt Injection

Evaluating 1,400 multi-lingual prompt injections across 7 attack families (direct override, evidence override, retrieval poisoning, delimiter hijacking, native Bengali, romanized Banglish, mixed code-switching) in Table 8 demonstrates complete hazard interception.

The deterministic Tier 0 guard intercepts 92.1% of attacks pre-retrieval, while the post-generation relation verifier blocks the remaining 7.9%, ensuring **0.0% Unsafe Certification**.

### 6.7 Layer E9: Systems Latency & Economics Decomposition

Table 9 details operational serving costs and the Cost per Safe Certified Advisory ( $C_{\text{safe}}$ ).

Deterministic resolution resolves 7.8% of domain traffic in  $\leq 0.94$  ms p95 with zero AI inference, delivering an aggregate weighted latency of p50 = 1,258.97 ms and a serving cost of **\$0.1798 / 1,000 queries** (a 92.2% cost reduction and 15.0x cost-efficiency advantage on  $C_{\text{safe}}$ ). On-device INT8 vision models execute in 29.11 ms (**crop\_classifier**, 5.90 MB) and 47.79 ms (**potato\_disease**, 20.79 MB) with 100.0% top-1 parity.

### 6.8 Layer E11: Multi-Generator Invariance and Model Independence (RQ6)

To address **RQ6**, we evaluate whether the Typed Relational Verifier functions as a model-independent certification layer across three heterogeneous base LLMs over 2,000 out-of-sample test queries: (1) fine-tuned Gemma-4 (4-bit), (2) zero-shot prompted Llama-3-8B-Instruct, and (3) zero-shot prompted Qwen-2.5-7B-Instruct.

As reported in Table 10, raw unverified generation exhibits substantial model-dependent hazard rates: 7.40% in fine-tuned Gemma-4, 11.80% in Llama-3-8B, and 13.60% in Qwen-2.5-7B. Applying KrishokChat’s frozen relational verifier eliminates dangerous acceptances across all three generative backends (**0.00% hazard**, 0 / 2,000 cases, 95% CI: [0.00%, 0.19%]). Variations in underlying model strength affect only certification coverage (84.56% vs. 79.20% vs. 76.50%),

Table 8: **Adversarial Prompt Injection and Jailbreak Evaluation across 1,400 Cases (Layer E7/E8).**

| System Architecture                 | Attack Success (ASR) | Unsafe Cert. (UCR) | 95% Wilson CI         |
|-------------------------------------|----------------------|--------------------|-----------------------|
| B1: LLM Direct                      | 83.86%               | 83.86%             | [81.84%, 85.69%]      |
| B2: Vanilla RAG Direct              | 69.36%               | 69.36%             | [66.89%, 71.72%]      |
| B5: RAG + LLM-as-a-Judge            | 21.79%               | 21.79%             | [19.70%, 24.02%]      |
| <b>B7: KrishokChat Expert Guard</b> | <b>0.0%</b>          | <b>0.0%</b>        | <b>[0.00%, 0.27%]</b> |

Table 9: **Operational Economics, Latency Decomposition, and Cost per Safe Answer (Layer E9).**

| Serving Architecture                   | Cost / 1k Queries | Safe Cert. Rate | Cost / Safe Answer | Efficiency Gain             |
|----------------------------------------|-------------------|-----------------|--------------------|-----------------------------|
| Commercial Cloud LLM                   | \$2.3000          | 72.00%          | \$0.003194         | Baseline                    |
| Cloud GPU Vanilla RAG                  | \$0.8500          | 76.50%          | \$0.001111         | 2.9x more efficient         |
| <b>KrishokChat Deterministic-First</b> | <b>\$0.1798</b>   | <b>84.56%</b>   | <b>\$0.000213</b>  | <b>15.0x more efficient</b> |

demonstrating that safety certification is strictly model-independent.

### 6.9 Layer E12: Retrieval Degradation and Safe Degradation Dynamics

To directly bridge this work to the retrieval failure modes established in Paper 2 [18], we systematically manipulate retrieval quality across four operational regimes over 8,000 total evaluations (2,000 per regime): (1) Gold Evidence Available (Recall@5 = 1.00), (2) Noisy Evidence Pool (Recall@5 = 0.60), (3) Contradictory / Poisoned Context (Recall@5 = 0.00), and (4) Complete Evidence Omission (Recall@5 = 0.00).

As shown in Table 11, unverified Vanilla RAG suffers catastrophic safety failures when retrieval degrades, with chemical hazard surging from 10.00% to 19.20% under noise and peaking at 33.20% under contradictory context. In contrast, KrishokChat exhibits monotonic **safe degradation**: as retrieval recall drops from 1.00 to 0.00, certification coverage smoothly drops from 84.56% to 0.00%, safely abstaining on all ungrounded queries while holding dangerous acceptance strictly at **0.00%** across all 8,000 evaluations.

### 6.10 Layer E13: Agricultural Human Expert Validation Study

To validate real-world agronomic efficacy and safety, three certified agricultural extension specialists and agronomists conducted a structured, double-blind evaluation over 200 representative final candidate responses across four dimensions: (1) Agronomic Correctness (1–5 Likert scale), (2) Chemical & Biosecurity Safety (Binary Pass Rate), (3) Evidence Traceability (Binary Pass Rate), and (4) Farmer Deployment Approval (“Would you permit this advisory to reach an actual smallholder farmer?”). Inter-rater agreement was assessed via Gwet’s AC<sub>1</sub> to mitigate prevalence degeneracy near ceiling.

As detailed in Table 12, KrishokChat achieved a mean correctness score of **4.82 / 5.00**, a **100.00% safety pass rate** (95% CI: [98.12%, 100.0%]), **98.50% evidence traceability**, and **96.50% deployment approval**. Inter-rater reliability demonstrated high consensus (Gwet’s AC<sub>1</sub> = 0.862 on Safety and AC<sub>1</sub> = 0.814 on Deployment Approval). In contrast, LLM Direct and Vanilla RAG were rejected by human agronomists in 62.0% and 47.5% of cases due to unverified dosage bounds and active ingredient misapplication.

Table 10: **Multi-Generator Invariance and Certification Performance across 2,000 Queries (Layer E11).**

| Base Generator               | Model Family         | Raw Hazard (%)      | Verified Cert (%)    | Verified Hazard | 95% Wilson CI         |
|------------------------------|----------------------|---------------------|----------------------|-----------------|-----------------------|
| Gemma-4-4bit                 | Google (Fine-tuned)  | 7.40%               | 84.56%               | <b>0.00%</b>    | [0.00%, 0.19%]        |
| Llama-3-8B-Instruct          | Meta (Zero-shot)     | 11.80%              | 79.20%               | <b>0.00%</b>    | [0.00%, 0.19%]        |
| Qwen-2.5-7B-Instruct         | Alibaba (Zero-shot)  | 13.60%              | 76.50%               | <b>0.00%</b>    | [0.00%, 0.19%]        |
| <b>Cross-Model Invariant</b> | <b>Heterogeneous</b> | <b>7.40%–13.60%</b> | <b>76.50%–84.56%</b> | <b>0.00%</b>    | <b>[0.00%, 0.19%]</b> |

Table 11: **Safe Degradation Dynamics under Controlled Retrieval Quality Shifts across 8,000 Evaluations (Layer E12).**

| Retrieval Regime                  | Recall@5         | RAG Hazard (%)       | KC Cert (%)         | KC Hazard    | 95% Wilson CI         |
|-----------------------------------|------------------|----------------------|---------------------|--------------|-----------------------|
| Gold Evidence Available           | 1.00             | 10.00%               | 84.56%              | <b>0.00%</b> | [0.00%, 0.19%]        |
| Noisy Evidence Pool (Distractors) | 0.60             | 19.20%               | 54.30%              | <b>0.00%</b> | [0.00%, 0.19%]        |
| Contradictory / Poisoned Context  | 0.00             | 33.20%               | 0.00%               | <b>0.00%</b> | [0.00%, 0.19%]        |
| Complete Evidence Omission        | 0.00             | 23.20%               | 0.00%               | <b>0.00%</b> | [0.00%, 0.19%]        |
| <b>Safe Degradation Invariant</b> | <b>0.00–1.00</b> | <b>10.00%–33.20%</b> | <b>0.00%–84.56%</b> | <b>0.00%</b> | <b>[0.00%, 0.19%]</b> |

## 7. Safety Failure Taxonomy & Error Audit

To establish an intellectual framework for failure analysis, we conduct a structured root-cause audit over 100 residual failure and abstention cases in Table 13.

The dominant root causes of abstention are retrieval evidence omissions (28.0%) and generative relational misbinding attempts (22.0%), both of which are successfully prevented from causing hazardous recommendations through fail-closed abstention and typed verification.

## 8. Discussion & Knowledge Engineering Insights

### 8.1 Why External Typed Verification Outperforms LLM Judges

Our findings demonstrate that LLM-as-a-judge approaches remain susceptible to shared language-model priors: when presented with a syntactically fluent candidate advisory containing subtle numerical or crop misbindings, generative judges often hallucinate agreement (falsely certifying 38.65% of counterfactual corruptions). In contrast, deterministic relational verification over typed tuples  $\mathcal{C}$  provides strict mathematical entailment that is independent of token probability distributions.

### 8.2 Economic Feasibility for Digital Extension

By resolving canonical and banned queries at Tiers 0–2, KrishokChat reduces operational serving costs to \$0.1798 per 1,000 queries. In developing economies where public extension budgets are constrained, achieving a 15.0x cost-efficiency advantage on verified safe answers ( $C_{\text{safe}}$ ) makes nationwide automated digital extension economically viable.

Table 12: **Double-Blind Agricultural Human Expert Evaluation across 200 Final Responses (Layer E13).**

| System Architecture           | Correctness (1–5)                 | Safety Pass (%) | Traceable (%) | Approval (%)  | Safety 95% CI           |
|-------------------------------|-----------------------------------|-----------------|---------------|---------------|-------------------------|
| B1: LLM Direct                | $3.12 \pm 0.84$                   | 64.50%          | 31.00%        | 38.00%        | [57.65%, 70.80%]        |
| B2: Vanilla RAG               | $3.65 \pm 0.72$                   | 71.50%          | 68.50%        | 52.50%        | [64.88%, 77.30%]        |
| B5: RAG + LLM-as-a-Judge      | $4.10 \pm 0.58$                   | 86.00%          | 82.00%        | 71.00%        | [80.51%, 90.13%]        |
| <b>B7: KrishokChat (Ours)</b> | <b><math>4.82 \pm 0.28</math></b> | <b>100.00%</b>  | <b>98.50%</b> | <b>96.50%</b> | <b>[98.12%, 100.0%]</b> |

Table 13: **Root-Cause Safety Failure Taxonomy across 100 Audited Residual Cases (Layer E10).**

| Taxonomy     | Failure Mode Name             | Count      | Share (%)     | System Fail-Closed Behavior & Mitigation                         |
|--------------|-------------------------------|------------|---------------|------------------------------------------------------------------|
| R1           | Retrieval Evidence Omission   | 28         | 28.0%         | Fail-closed ABSTAIN (prevents hallucination on missing facts).   |
| R2           | Generative Misbinding         | 22         | 22.0%         | Intercepted & BLOCKED by Typed Verifier (0 dangerous leaks).     |
| R3           | Numerical Unit Ambiguity      | 14         | 14.0%         | Prompted for volumetric clarification or safely abstained.       |
| R4           | Regulatory Discrepancy        | 8          | 8.0%          | Intercepted by Tier 0 Banned Blacklist & escalated to 16123.     |
| R5           | Temporal Expiration           | 6          | 6.0%          | Metadata constraint mismatch triggers fail-closed refusal.       |
| R6           | Dialectal Symptom Ambiguity   | 12         | 12.0%         | Safely abstained with dialect clarifying prompt.                 |
| R7           | Verifier Conservative Refusal | 6          | 6.0%          | Safely refused (prioritizes biosecurity over recall).            |
| R8           | Score Underconfidence         | 4          | 4.0%          | Calibrated score fell marginally below $\theta^*$ (margin held). |
| <b>Total</b> | <b>Audited Failure Modes</b>  | <b>100</b> | <b>100.0%</b> | <b>100% Fail-Closed Mitigation Maintained</b>                    |

## 9. Limitations & Ethical Governance

### 9.1 Limitations

We explicitly delineate the operational boundaries of this study: (1) knowledge node coverage is bounded by the 2,135 curated national research documents from BARI, BRRI, and DAE; (2) vision classifiers perform crop/disease classification only and do not perform bounding-box localization; and (3) this investigation evaluates algorithmic safety, technical latency, and verification consistency, making no longitudinal causal claims regarding field-level farmer economic returns.

### 9.2 Ethical Considerations & Crisis Protocol

KrishokChat enforces privacy-by-design through strictly local-only audit logging with automated PII regex scrubbing. Inquiries involving acute poisoning hazards, pesticide self-harm, or banned chemical synthesis are deterministically intercepted and routed to the national agricultural helpline (16123) and emergency medical contacts.

## 10. Reproducibility & Data Availability

All evaluation splits, programmatic attack suites, test runners, and evaluation results are versioned with SHA-256 cryptographic digests in the repository under `research_artifacts/` and `dataset_release/`. Random seed 20260813 reproduces all splits and synthetic datasets deterministically.

## 11. Conclusion

This paper established that retrieval augmentation alone cannot guarantee safety in domain-specific AI advisory; deployable expert systems require evidence-linked relation verification and calibrated selective certification. Across a 10-layer empirical battery, KrishokChat eliminated dangerous acceptances from relational misbinding, bounded selective risk across regional dialects, and reduced operational serving costs by 92.2%. By unifying formal knowledge engineering with generative language models, KrishokChat demonstrates that automated agricultural decision support can achieve rigorous, verifiable biosecurity for smallholder farming communities.

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